DATA ANALYST INTERNSHIP — TASK 5

Exploratory Data Analysis (EDA): Titanic Dataset

Objective

To explore a dataset using statistical summaries and visualizations, identify patterns and trends, inspect data quality, and communicate findings.

Dataset and tools

Dataset: Titanic passenger training dataset (commonly distributed as 891 passenger records). Tools: Python, Pandas, NumPy, Matplotlib, and Seaborn. The accompanying notebook loads the CSV from a public GitHub source.

1. Data understanding

The analysis begins with the dataset dimensions, column data types, descriptive statistics, and frequency counts. These steps help identify numerical and categorical variables and understand their ranges and distributions.

2. Data quality and preparation

Missing values are inspected before cleaning. The notebook fills missing Age values with the median and missing Embarked values with the mode. It also creates a HasCabin indicator to show whether a cabin value was recorded. These are simple introductory choices; other projects may require different treatment.

3. Univariate analysis

Histograms are used to inspect numerical variables such as Age and Fare. Count plots summarize categories such as Sex, Pclass, Embarked, and Survived. In the standard Titanic training data, Fare is strongly right-skewed, and third-class passengers make up the largest passenger class.

4. Bivariate analysis

Bar plots and cross-tabulations compare survival with sex and passenger class. In the standard Titanic training dataset, female passengers had a substantially higher survival rate than male passengers. Survival was also higher in first class than in third class. These are observed associations and do not establish causation.

5. Multivariate analysis and correlation

A correlation heatmap summarizes pairwise linear relationships among numeric variables. A pairplot helps inspect relationships among selected variables. Correlation values describe association, not causation; categorical variables and non-linear relationships may require additional methods.

6. Outlier analysis

Boxplots and the IQR rule are used to flag potential outliers in Age and Fare. A flagged value is not automatically an error. High fares may reflect genuine differences in ticket pricing, so outliers should be investigated before removal.

Conclusion

The Titanic dataset demonstrates how EDA can reveal differences in survival across passenger groups, identify skewed numerical variables, and highlight missing values and potential outliers. Sex and passenger class show clear relationships with survival in this dataset. The notebook contains the reproducible code and visualizations used to examine these patterns.

Important submission note

Run all cells in the supplied notebook before submitting so that your notebook contains the actual outputs and plots. If your downloaded CSV differs from the standard version, update the report observations to match your results. The task brief asks for a Jupyter Notebook and a PDF report, and also recommends adding code, dataset, screenshots (if any), and a short README to a new GitHub repository.